

|           |          |                                                                                                                                                                                     |
|-----------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|           |          | flow from M towards N. Hence, N will be at the higher potential.                                                                                                                    |
| <b>26</b> | <b>B</b> | Peak voltage in secondary coil is 90 V.<br>Resistor $R_2$ and $R_3$ are in parallel connection,<br>so total resistance in circuit connected to the secondary coil is 120 $\Omega$ . |

| Qn | Ans | Suggested Solution                                                                                                                                                                                                                                                                                                                                           |
|----|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    |     | <p>R.m.s. voltage in secondary coil, <math>V_{\text{r.m.s.}(2)} = \frac{90}{\sqrt{2}} \text{ V}</math></p> <p>R.m.s. current in secondary coil, <math>I_{\text{r.m.s.}(2)} = \frac{\frac{90}{\sqrt{2}}}{120} \text{ A}</math></p> <p>Average power dissipated in <math>R_1</math></p> $= (I_{\text{r.m.s.}(2)})^2 \times 80$ $= 22.5$ $\approx 23 \text{ W}$ |
| 27 | A   | 2                                                                                                                                                                                                                                                                                                                                                            |